

Nginx Deployment Lab Manual - Two Approaches

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This lab teaches two methods of deploying a simple website using Nginx.

Approach 1: Use the official Nginx Docker image.

Approach 2: Install Nginx manually inside an Ubuntu Linux container.

Each exercise contains:

- Objective
- Command
- Explanation
- Expected Output
- Screenshot Placeholder

Approach 1 - Pull Nginx Image

Command:

```
docker pull nginx
```

Explanation:

Downloads the official Nginx image from Docker Hub.

Expected Output: Pull Nginx image of latest version from Docker Hub

Screenshot Placeholder

```
dhharshinik@DHC47TYX72-dharshinik docker % docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
78e1a52d6c50: Pull complete
43899da09563: Pull complete
97a97a349235: Pull complete
0ef4315fa7b7: Pull complete
572aad3b769d: Pull complete
8ae94b355a63: Pull complete
3c03b4506f3b: Download complete
37647833f672: Download complete
Digest: sha256:42f2d24ae18df9b5251d1cc45548085656d2335e9338fd150a24e415462d151f
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview nginx
dhharshinik@DHC47TYX72-dharshinik docker % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	28bd5fe8b56d	13.6MB	4.27MB	U
banking-api:1.0	fdabb506d5ef	393MB	98.5MB	U
banking.api:1.0	ae039ac8854d	393MB	98.5MB	
compose-postgres-api-lab-api:latest	b880a79d9b55	387MB	97MB	U
nginx:latest	42f2d24ae18d	259MB	64.3MB	

Approach 1 - Run Nginx Container

Command:

```
docker run -d --name myweb -p 8080:80 nginx
```

Explanation:

Starts an Nginx container and maps port 8080 on the host to port 80 inside the container.

Expected Output: Create and run nginx image based container in

- -d : detach mode
- -p: mentioned port
- --name : name of the container
- 8080 => host machine port
- 80 => default container port

Screenshot Placeholder

```
myweb
dharshinik@DHC47TYX72-dharshinik Nginx % docker run -d --name myweb -p 8080:80 nginx
d2cde6457a05950646e31353851ae29a1a13d7d72266d8cee543226b433c45a
dharshinik@DHC47TYX72-dharshinik Nginx % docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
d2cde6457a05   nginx    "/docker-entrypoint..." 4 seconds ago Up 3 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp myweb
dharshinik@DHC47TYX72-dharshinik Nginx %
```

Approach 1 - Verify Running Container

Command:

```
docker ps
```

Explanation:

Displays running containers.

Expected Output: use docker process status (ps) to view containers in running state

Screenshot Placeholder

```

dharshinik@DHC47TYX72-dharshinik docker % docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
89f726d4063a   nginx    "/docker-entrypoint..." 2 seconds ago    Up 2 seconds    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp    myweb
dharshinik@DHC47TYX72-dharshinik docker %
```

Approach 1 - Create Custom index.html

Command:

Create a local index.html file.

Explanation:

Creates a custom web page.

Expected Output: Create a .html file using touch command

Screenshot Placeholder

```
● dharshinik@DHC47TYX72-dharshinik Nginx % touch index.html
● dharshinik@DHC47TYX72-dharshinik Nginx % ls
index.html
○ dharshinik@DHC47TYX72-dharshinik Nginx %
```

```
1 <!--DOCTYPE html-->
2 <html>
3 <head>
4 |   <title>Nginx</title>
5 </head>
6 <body>
7 |   <h1>Hello, World!</h1>
8 |   <p>Hosting webpages in Nginx.</p>
9 </body>
10 </html>
```

Approach 1 - Copy File

Command:

```
docker cp index.html myweb:/usr/share/nginx/html/index.html
```

Explanation:

Copies the local HTML file into the Nginx web root.

Expected Output: copies index.html to myweb container

- Command breakdown : docker cp source destination : destinationLocation

Screenshot Placeholder



```
dharshtnik@DHC47TYX72-dharshhtnik Nginx % docker cp index.html myweb:/usr/share/nginx/html/index.html
Successfully copied 150B (transferred 2.05kB) to myweb:/usr/share/nginx/html/index.html
```

Approach 1 - Verify Website

Command:

Browse to `http://localhost:8080`

Explanation:

Displays the custom web page.

Expected Output: The container is running, so the website will be hosted

Screenshot Placeholder

Hello, World!

Hosting webpages in Nginx.

```
● dharshinik@DHC47TYX72-dharshinik Nginx % curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
  <title>Nginx</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>Hosting webpages in Nginx.</p>
</body>
</html>
● dharshinik@DHC47TYX72-dharshinik Nginx %
```

Approach 2 - Pull Ubuntu

Command:

```
docker pull ubuntu:24.04
```

Explanation:

Downloads the Ubuntu Linux image.

Expected Output: pull the ubuntu image from DockerHub

Screenshot Placeholder

```
dharshinik@DHC47TYX72-dharshinik docker % docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:53958ec7b67c2c9355df922dd08dbf0360611f8c3cdb656875e81873db9ffdba
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```


Approach 2 - Start Ubuntu

Command:

```
docker run -it --name ubuntu-nginx ubuntu:24.04 bash
```

Explanation:

Starts an interactive Ubuntu container.

Expected Output: creates and runs a nginx based container in bash terminal

Screenshot Placeholder

A terminal window with a dark background. The prompt is 'dharshinik@DHC47TYX72-dharshinik docker %'. The command entered is 'docker run -it ubuntu bash'. The output shows the user is now 'root@2cade0d6eaf7:/#' with a cursor at the end of the line.

```
dharshinik@DHC47TYX72-dharshinik docker % docker run -it ubuntu bash
root@2cade0d6eaf7:/#
```

Approach 2 - Update Package Repository

Command:

apt update

Explanation:

Downloads the latest package metadata.

Expected Output: The latest package metadata is installed in package configuration

Screenshot Placeholder

```
root@2cade0d6eaf7:/# apt update
Get:1 http://ports.ubuntu.com/ubuntu-ports resolute InRelease [136 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports resolute-updates InRelease [137 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports resolute-backports InRelease [136 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports resolute-security InRelease [137 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports resolute/main arm64 Packages [1860 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports resolute/restricted arm64 Packages [231 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports resolute/universe arm64 Packages [19.9 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports resolute/multiverse arm64 Packages [298 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports resolute-updates/main arm64 Packages [318 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports resolute-updates/restricted arm64 Packages [328 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports resolute-updates/universe arm64 Packages [178 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports resolute-updates/multiverse arm64 Packages [3592 B]
Get:13 http://ports.ubuntu.com/ubuntu-ports resolute-security/main arm64 Packages [270 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports resolute-security/universe arm64 Packages [147 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports resolute-security/restricted arm64 Packages [324 kB]
Fetched 24.4 MB in 9s (2849 kB/s)
All packages are up to date.
```

Approach 2 - Install Nginx

Command:

```
apt install nginx -y
```

Explanation:

Installs the Nginx web server.

Expected Output: downloads nginx within the linux container (-y = yes to all confirmation prompts of download)

Screenshot Placeholder

```
root@3b36ca549ed6:/# apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 646 kB
  Space needed: 1866 kB / 448 GB available
```

Approach 2 - Verify Nginx

Command:

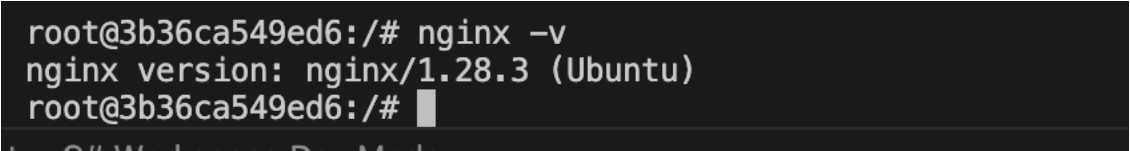
```
nginx -v
```

Explanation:

Displays the installed Nginx version.

Expected Output: Tells the downloaded version of nginx package

Screenshot Placeholder



```
root@3b36ca549ed6:/# nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
root@3b36ca549ed6:/#
```

Approach 2 - Create Custom Page

Command:

Edit /var/www/html/index.html

Explanation:

Creates a custom website page.

Expected Output: Create index.html with touch command.

Screenshot Placeholder

```
root@3b36ca549ed6:/# touch /var/www/html/index.html
root@3b36ca549ed6:/# ls
bin boot dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var
root@3b36ca549ed6:/# cd var
root@3b36ca549ed6:/var# cd www
root@3b36ca549ed6:/var/www# cd html
root@3b36ca549ed6:/var/www/html# ls
index.html index.nginx-debian.html
root@3b36ca549ed6:/var/www/html#
```

```
root@3b36ca549ed6:/var/www/html# echo "<h1>html file content from Linux terminal based nginx hosting</h1>" > index.html
root@3b36ca549ed6:/var/www/html# cat index.html
<h1>html file content from Linux terminal based nginx hosting</h1>
root@3b36ca549ed6:/var/www/html#
```

Approach 2 - Start Nginx

Command:

```
nginx
```

Explanation:

Starts the Nginx service.

Expected Output: to start nginx server

Screenshot Placeholder



```
root@3b36ca549ed6:/var/www/html# nginx
```

Approach 2 - Verify Locally

Command:

```
curl localhost
```

Explanation:

Requests the page from the web server.

Expected Output: Check the endpoint

Screenshot Placeholder

```
root@3b36ca549ed6:/var/www/html# curl localhost
<h1>html file content from Linux terminal based nginx hosting</h1>
root@3b36ca549ed6:/var/www/html#
```

Approach 2 - Commit Container

Command:

```
docker commit ubuntu-nginx ubuntu-nginx:v1
```

Explanation: Saves the modified container as a reusable image.

Expected Output: saves the modified container to new image

- ubuntu-nginx is the container name
- ubuntu-nginx:v1 is the image name

Screenshot Placeholder

```
pg
● dharshinik@DHC47TYX72-dharshinik docker % docker commit funny_swanson ubuntu-nginx:v1
sha256:ef5ef3988df11dd2af440a1223e0645b108a06bd9ac2cc0681584c684feda877
● dharshinik@DHC47TYX72-dharshinik docker % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	28bd5fe8b56d	13.6MB	4.27MB	U
banking-api:1.0	fdabb506d5ef	393MB	98.5MB	U
banking.api:1.0	ae039ac8854d	393MB	98.5MB	
compose-postgres-api-lab-api:latest	b880a79d9b55	387MB	97MB	U
nginx:latest	42f2d24ae18d	259MB	64.3MB	U
postgres:16	081f1bc7bd5e	663MB	165MB	U
postgres:latest	29ee7bb30d80	671MB	167MB	U
ubuntu-nginx:v1	ef5ef3988df1	278MB	77.1MB	
ubuntu:latest	53958ec7b67c	180MB	44.4MB	U
weatherapi:1.0	0c50d191dbab	352MB	88.9MB	
weatherapi:single	a13326415044	1.3GB	320MB	

Approach 2 - Run Published Website

Command:

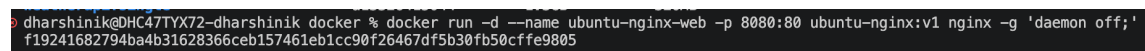
```
docker run -d --name ubuntu-nginx-web -p 8080:80 ubuntu-nginx:v1 nginx -g 'daemon off;'
```

Explanation:

Runs the custom image and exposes the website.

Expected Output: Creates a container ubuntu-nginx-web of the image: ubuntu-nginx:v1 in attach mode ('daemon off')

Screenshot Placeholder



```
dharshinik@DHC47TYX72-dharshinik docker % docker run -d --name ubuntu-nginx-web -p 8080:80 ubuntu-nginx:v1 nginx -g 'daemon off;'  
f19241682794ba4b31628366ceb157461eb1cc90f26467df5b30fb50cffe9805
```